

Pseudo-Last Design for SUPR-Guided Shoe Reconstruction

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1 Pseudo-Last Design for SUPR-Guided Shoe Reconstruction

1.1 Design target

For your use case, the right target is **not** an industrial manufacturing last and **not** a literal copy of the SUPR foot. It is a **computational, watertight, smooth interior prior** that preserves the hidden sole/interior support region well enough to regularize a NeuS-style or G-Shell-style reconstruction when the shoe opening hides much of the inner topology. That framing matches the literature on both sides of your problem: NeuS reconstructs a zero-level SDF from images but notes that severe self-occlusion and thin structures remain difficult without stronger geometric constraints, while G-Shell explicitly represents open surfaces as islands on a watertight shell using a manifold SDF, making a watertight interior-support prior especially natural for the hidden parts of a shoe. SUPR is also a better anatomical seed than older generic body models because it explicitly improves feet, toe motion, and foot deformation under ground contact using large-scale body-part and 4D foot scans.

In digital footwear work, the shoe last is treated as the core determinant of shoe shape, fit, and volume, and custom-last workflows typically align the foot to a heel-based axis, then apply **region-specific ease allowances** and **cross-sectional editing** rather than copying raw anatomy vertex-for-vertex. That is exactly the right design philosophy here: treat the SUPR foot as the anatomical low-frequency prior, and build a smoother, more shoe-like pseudo-last that is intentionally biased toward stable hidden topology.

My recommendation is therefore:

Use a section-based pseudo-last pipeline.

Preserve the plantar support geometry and overall hindfoot/forefoot volume cues from the aligned SUPR foot, but replace the literal dorsal/toe anatomy with a smoother last-like upper envelope, then convert that mesh into a watertight SDF/occupancy prior for shell and mSDF supervision.

1.2 Why a last is not a foot

A real shoe last is a **rigid shaping tool**, not a literal foot twin. Footwear references and last-design papers describe the last as the object that determines shoe style, fit, and internal volume, and last-processing papers explicitly note that a scanned foot must typically be **adjusted** before it matches last conventions because footwear includes features such as toe spring, heel height, and intentionally distributed ease. In other words, the last is already a geometric abstraction of the foot, with added design bias.

A foot, by contrast, is articulated, compliant, and dynamically changing. Dynamic foot-shape studies show that foot morphology changes under loading; safety-shoe and walking studies report practically relevant changes in width and girth between static and dynamic conditions, and 4D statistical models capture both inter-person and within-gait variation. That matters because copying the bare anatomical foot too faithfully will overfit the wrong signal in exactly the regions that should instead become a smoother shoe cavity.

The forefoot makes this especially clear. Toe-box studies show that shoe toe-box shape and volume materially affect dorsal and plantar forefoot pressures, with sharper or more constrained toe regions increasing pressure in sensitive areas. Put bluntly: **individual toes are anatomically real, but they are not the right topology for a sneaker toe-box prior.**

A final point is population variation. Large-scale foot-scan analyses and footwear-focused reviews show substantial differences across sex and region, and the review literature is explicit that women's feet should not be treated as scaled-down men's feet. For your problem, the implication is not that you need sex-specific last engineering on day one, but that a good pseudo-last should be **parametric and smoothable**, not hard-coded around one raw foot surface.

1.3 Anatomy to preserve and anatomy to modify

The simplest useful rule is this:

Preserve low-frequency, support-relevant anatomy. Smooth high-frequency, shoe-hostile anatomy.

The parts I would preserve are the **plantar heel, ball region, general forefoot spread, overall instep dome, heel bulb / calcaneal back volume**, and a **moderated medial arch uplift**. Those are the features most closely tied to support, contact, and functional fit in the footwear and orthotic literature, and they are exactly the kinds of landmarks and measurements most foot-shape pipelines focus on: heel point, toe point, ball landmarks, arch region, dorsal height, midfoot width, and girths. Plantar-shape studies also show that weight-bearing plantar geometry matters for support design, so preserving bottom shape in some form is important.

The parts I would deliberately smooth or remove are the **shank/leg above the ankle, separate toes, sharp metatarsal-head bumps, malleolar/ankle prominences**, and any small dorsum/navicular protrusions that create local waviness without adding useful support semantics

for a closed sneaker. That choice is consistent with section-based last workflows and with ease-allowance approaches that distribute fit volume smoothly across last regions rather than preserving every anatomical protrusion.

So, in practical terms:

- **Keep:** plantar support shape, hindfoot cup volume, forefoot width trend, instep height trend.
- **Soften:** arch details into a clean midfoot sweep.
- **Merge:** toes into one toe-box mass.
- **Tighten:** upper heel slightly for hold.
- **Delete:** anything above the intended collar/opening zone.

That gives you a prior that is anatomically anchored where it matters and topologically conservative where image supervision is weak.

1.4 Mesh construction algorithm

1.4.1 Preprocess and crop the SUPR foot

Use the aligned neutral SUPR right foot as the starting mesh M_f . Because your shoes are already normalized, operate entirely in normalized coordinates and define the foot length

$$L_f = x_{\max}(M_f) - x_{\min}(M_f).$$

I recommend estimating a **leg axis** from the rear-top vertex cluster rather than using a naive axis-aligned clip. If SUPR joint data are available, use the ankle joint and local shank axis directly; otherwise, estimate the axis by PCA over vertices in the rearfoot and upper-shank cluster. Then sweep a family of planes orthogonal to that axis near the ankle and pick the plane whose section perimeter is minimal. This removes the leg above the ankle much more cleanly than clipping by $x = \text{const}$. That is consistent with footwear alignment literature, which emphasizes robust reference axes and warns that simple heel-to-toe alignment can be unstable when toe geometry is irregular.

After that crop, keep the foot-side component and triangulate a temporary cap over the cut boundary. This temporary cap is only for downstream section extraction and SDF cleanup; it does not need to be final-quality.

1.4.2 Build a shoe-constrained centerline and width profile

Even if you already have a support centerline, I would recompute a **sectional centerline** from the support footprint as a consistency check.

At each slice x_i , intersect the support footprint with the line $x = x_i$. Let the largest connected interval be

$$I_i = [z_L(x_i), z_R(x_i)].$$

Define the raw center and support width as

$$c_i^0 = \frac{z_L(x_i) + z_R(x_i)}{2}, \quad w_i^0 = z_R(x_i) - z_L(x_i).$$

Then smooth both with cubic smoothing splines or a discrete second-derivative regularizer:

$$c = \arg \min_{\tilde{c}} \sum_i (\tilde{c}(x_i) - c_i^0)^2 + \lambda_c \sum_i (\Delta^2 \tilde{c}_i)^2,$$

$$w = \arg \min_{\tilde{w}} \sum_i (\tilde{w}(x_i) - w_i^0)^2 + \lambda_w \sum_i (\Delta^2 \tilde{w}_i)^2.$$

Because your shoes are already in a heel-to-toe coordinate frame, this x-monotone formulation is simpler and more stable than a full medial-axis skeleton for normal sneakers. Foot-measurement and retail-scan pipelines use essentially this kind of axis/outline logic when extracting robust length, width, and heel-width measurements.

1.4.3 Preserve the plantar shape using the footbed and the SUPR arch

Let $B(x, z)$ be your pseudo-footbed surface in the given coordinates, where larger y means more downward/closer to the sole. Let $y_f^{pl}(x, z)$ be the plantar surface of the cropped SUPR foot after local interpolation.

You do **not** want to flatten the pseudo-last bottom to the footbed everywhere, because that erases useful arch information. But you also do **not** want to use the bare foot plantar surface directly, because that may float unrealistically relative to the shoe support geometry.

A useful blend is:

$$y_b(x, z) = B(x, z) - \rho_b(s) [B(x, z) - y_f^{pl}(x, z)]_+, \quad s = \frac{x - x_h}{L_f},$$

where $[u]_+ = \max(u, 0)$, $x_h = x_{\min}(M_f)$, and $\rho_b(s)$ is an arch-preservation weight that is strong in the midfoot and weak in heel/toe support zones.

A good default is a Gaussian-like arch weight:

$$\rho_b(s) = \rho_{\text{arch}} \exp\left(-\frac{(s - 0.45)^2}{2(0.12)^2}\right).$$

This gives you a bottom surface that follows the shoe-derived footbed in supported regions while retaining moderated arch lift from the anatomical foot in the midfoot. Plantar-shape studies under different weight-bearing conditions strongly support the idea that the plantar surface should be treated as a controlled design variable in anything support-related.

1.4.4 Extract cross-sections and merge the toes

Now slice the cropped foot mesh at N_x equally spaced x -planes. Equal-interval sectioning is standard in digital last design; classic foot-to-last work even describes 1% foot-length sections, and more recent CAD work still builds shoe lasts from regular cross-sections and characteristic points.

For each slice x_i :

1. Intersect the mesh with the plane $x = x_i$.
2. Convert the resulting contour to local slice coordinates (ξ, h) , where

$$\xi = z - c(x_i), \quad h = y_b(x_i, z) - y.$$

Here h is height above the last bottom.

3. Rasterize the slice to a binary mask $M_i(\xi, h)$.

For the toe region, do **not** use the raw mask directly. Instead, starting at a normalized longitudinal position s_{merge} , apply morphological closing with an elliptical kernel whose radii increase toward the toe tip:

$$M'_i = (M_i \oplus K_i) \ominus K_i.$$

This closes the inter-toe gaps and turns the literal toe anatomy into one continuous toe-box section. It is the simplest robust way to enforce “one toe-box volume” without hand-sculpting. After closing, extract the contour of M'_i and fit a smooth superellipse or resampled B-spline.

1.4.5 Add toe allowance, heel hold, and optional side/top inflation

Compute a smoothed raw foot half-width $a_f(x)$ and raw foot height $H_f(x)$ from the post-processed slice masks, then regularize them along x with the same second-derivative smoothing used for the centerline.

Then define target allowances:

$$\delta_x = \alpha_t L_f$$

for toe length,

$$\delta_w(s) = W_f (\alpha_{mid} g(s; 0.30, 0.55) + \alpha_{fore} g(s; 0.60, 0.85))$$

for side clearance, and

$$\delta_h(s) = H_f(s) (\beta_{vamp} g(s; 0.35, 0.65) + \beta_{toe} g(s; 0.72, 0.95))$$

for top clearance, where $g(s; a, b)$ is a smoothstep profile.

Then clamp the width to the shoe support geometry:

$$a_t(x) = \min \left(\eta_s \frac{w(x)}{2}, a_f(x) + \delta_w(s) \right),$$

$$H(x) = H_f(x) + \delta_h(s).$$

The factor $\eta_s < 1$ is important when your support profile comes from an outer shoe mesh rather than the true inner cavity.

For the upper envelope, use a superellipse section:

$$H_{\text{shoe}}(x, \xi) = H(x) \left[1 - \left| \frac{\xi}{a_t(x)} \right|^{p(s)} \right]_+^{1/q(s)},$$

$$y_u(x, z) = y_b(x, z) - H_{\text{shoe}}(x, z - c(x)).$$

Then add a **height-dependent heel-hold contraction** in the hindfoot by shrinking the slice laterally after section generation:

$$z \leftarrow c(x) + (z - c(x)) \left(1 - \gamma_h(s) \tau^\nu \right), \quad \tau = \frac{y_b(x, z) - y}{H(x)},$$

with $\gamma_h(s)$ nonzero only in the hindfoot. This keeps the plantar heel width stable while tightening the upper heel volume, which is much closer to what a sneaker backpart actually needs.

For toe allowance, append a short extension cap of length δ_x beyond the anatomical toe tip and taper width/height smoothly to zero, for example with a hemispheroidal taper:

$$a_t(x_t + u) = a_t(x_t) \sqrt{1 - (u/\delta_x)^2}, \quad H(x_t + u) = H(x_t) \sqrt{1 - (u/\delta_x)^2}, \quad u \in [0, \delta_x].$$

1.4.6 Loft, cap, remesh, and do an SDF cleanup round-trip

After generating all contours:

- resample every contour to the same point count N_θ ,
- loft adjacent slices with a strip of triangles,
- triangulate the toe cap,
- triangulate the ankle-opening cap from the crop contour,
- isotropically remesh,
- smooth only non-plantar vertices,
- then do one **SDF round-trip**: mesh $\rightarrow$ signed grid $\rightarrow$ Marching Cubes.

That last step is the easiest way to remove small self-intersections and guarantee a watertight output. Because this pseudo-last is computational, not manufacturing-grade, the SDF round-trip is a feature, not a bug.

1.4.7 Reference pseudocode

```
def build_pseudolast(
    supr_mesh,
    foot_transform,
    footbed_B,          # callable B(x, z)
    support_footprint,  # x-z mask or interval function
    N_x=128,
    N_theta=128
):
    M = transform_mesh(supr_mesh, foot_transform)

    # --- crop leg above ankle ---
    leg_axis = estimate_leg_axis_pca(M)
    crop_plane = choose_min_perimeter_ankle_plane(M, leg_axis)
    M = cut_keep_foot_side(M, crop_plane)

    # --- support centerline / width ---
    xs = np.linspace(xmin(M), xmax(M), N_x)
    c_raw, w_raw = sectional_centerline_width(support_footprint, xs)
    c = smooth_profile(c_raw, lam=1e-3)
    w = smooth_profile(w_raw, lam=1e-3)

    # --- plantar bottom from footbed + moderated arch preservation ---
    y_bottom = build_bottom_surface(M, footbed_B, c, arch_strength=0.75)

    # --- extract and clean sections ---
    sections = []
    for x in xs:
        contour = mesh_section(M, x)
        mask = rasterize_section(contour, center_z=c(x), bottom=y_bottom)
        s = (x - xs[0]) / (xs[-1] - xs[0])

        if s >= 0.74:
            mask = morphological_close(mask, kernel=toe_kernel(s))
```

```

a_f, H_f = fit_section_stats(mask)
a_t, H_t = apply_clearances(
    a_f=a_f,
    H_f=H_f,
    support_half_width=0.93 * 0.5 * w(x),
    s=s
)

contour = superellipse_contour(a_t, H_t, p=section_p(s), q=section_q(s))
contour = apply_heel_hold(contour, s=s, shrink=0.05)
contour = map_section_to_world(contour, x=x, center_z=c(x), bottom=y_bottom)
sections.append(contour)

sections = add_toe_extension_cap(sections, toe_allowance=0.05 * foot_length(M))
mesh = loft_sections(sections)
mesh = cap_openings(mesh) # ankle cap + remaining caps
mesh = smooth_nonplantar(mesh)
mesh = sdf_roundtrip(mesh, resolution=192)

return mesh

```

1.5 Mathematical formulation and normalized defaults

1.5.1 Coordinates and section model

Use your coordinate convention directly:

- $+X$: heel $\rightarrow$ toe
- $+Y$: sole/down
- $-Y$: opening/up
- $+Z$: width

Define the normalized longitudinal coordinate

$$s = \frac{x - x_h}{L_f}, \quad L_f = x_t - x_h.$$

Define height above the bottom reference using your footbed:

$$h(x, y, z) = y_b(x, z) - y,$$

where y_b is the blended plantar surface from the previous section.

With this convention, the pseudo-last volume can be represented slice-wise as

$$\Omega_{\text{last}} = \left\{ (x, y, z) : x \in [x_h, x_t + \delta_x], y_u(x, z) \leq y \leq y_b(x, z) \right\},$$

and y_u is the smoother shoe-like upper envelope built from section width and height profiles.

The profiles that matter most are:

- centerline $c(x)$,
- half-width $a_t(x)$,
- section height $H(x)$,
- arch-preservation weight $\rho_b(s)$,
- heel-hold strength $\gamma_h(s)$.

All of them should be regularized along x by curvature penalties, not just smoothed ad hoc. In practice, the discrete objective

$$f^{sm} = \arg \min_f \sum_i (f_i - \hat{f}_i)^2 + \lambda \sum_i (\Delta^2 f_i)^2$$

works well for $f \in \{c, w, a_f, H_f\}$.

1.5.2 Recommended normalized priors

Because your meshes are normalized, I would treat all defaults as **dimensionless fractions** of foot length, foot width, or local sectional height. The toe-length numbers below are anchored to footwear-fit literature that commonly recommends about 10-15 mm of extra length for adults; the width-related values are engineering priors informed by the literature on dynamic foot widening, pliable fit, and the fact that footwear fit still lacks one universal quantitative standard.

Quantity	Symbol	Recommended range	Good default
Toe length allowance	α_t	$0.040L_f$ to $0.060L_f$	$0.050L_f$
Toe merge start	s_{merge}	0.72 to 0.78	0.74
Full toe-box by	s_{box}	0.86 to 0.92	0.88
Midfoot side clearance	α_{mid}	$0.010W_f$ to $0.030W_f$	$0.020W_f$
Forefoot side clearance	α_{fore}	$0.030W_f$ to $0.060W_f$	$0.040W_f$
Vamp / top clearance	β_{vamp}	0.04 to 0.08 of local height	0.05
Toe vertical clearance	β_{toe}	0.03 to 0.07 of local height	0.04
Heel clearance	α_{heel}	0.01 to 0.03 of local heel width	0.015
Heel-hold upper shrink	γ_h	0.03 to 0.08 of hindfoot upper width	0.05
Arch/bottom preservation	ρ_{arch}	0.60 to 0.90	0.75
Support-to-inner width scale	η_s	0.90 to 0.96	0.93
Forefoot superellipse exponent	p	3.0 to 4.5	3.5
Vertical superellipse exponent	q	1.8 to 2.8	2.2

A few practical notes matter here.

First, **toe allowance should be longer than side allowance is wide**. If you inflate width aggressively but barely extend length, the pseudo-last becomes bulbous rather than sneaker-like.

Second, **heel clearance should stay meaningfully smaller than forefoot side clearance**. Otherwise the prior will teach the shell that heel looseness is acceptable, which is usually the opposite of what stabilizes topology in a low-top shoe.

Third, if your support width was extracted from the **outer** shoe mesh, start with $\eta_s \approx 0.93$. If it came from a cleaner estimate of the footbed/interior support region, you can move it toward 0.96.

1.6 SDF conversion and mSDF-guided training

1.6.1 Convert the pseudo-last mesh to an SDF or occupancy field

Once you have a watertight pseudo-last mesh M_L , convert it to a continuous field on the same domain as your reconstruction model.

For any query point p , define the unsigned distance

$$d_u(p) = \min_{t \in M_L} \|p - \Pi_t(p)\|_2,$$

where $\Pi_t(p)$ is the closest point on triangle t . For the sign, if the mesh is watertight, ray parity is fine; if you want robustness to small imperfections after lofting, use a generalized or fast winding-number test. That is a standard geometry-processing choice and works well for voxelization and signing distances.

Then define

$$d_L(p) = \text{sign}(p) d_u(p),$$

and a truncated field

$$\tilde{d}_L(p) = \text{clip}(d_L(p), -\tau, \tau), \quad \tau \in [0.03L_f, 0.06L_f].$$

If you need occupancy instead of SDF, use either

$$o_L(p) = \mathbf{1}[d_L(p) \leq 0]$$

or a differentiable soft occupancy

$$\hat{o}_L(p) = \sigma\left(-\frac{d_L(p)}{\sigma_o}\right), \quad \sigma_o \approx 0.01L_f.$$

DeepSDF and Occupancy Networks are still the right conceptual references here: continuous SDFs give you a boundary plus inside/outside semantics, while occupancy fields give you a smooth classifier at arbitrary resolution.

1.6.2 Use the pseudo-last in shell training first

For G-Shell-style training, I would use the pseudo-last in **two stages**.

The first stage regularizes the **watertight shell SDF** d_θ . Assume the usual sign convention that $d_\theta(p) < 0$ means the point lies inside the current shell volume.

Sample points on the pseudo-last surface S_L , and enforce that the pseudo-last sits strictly inside the learned shell by a margin m_{in} :

$$\mathcal{L}_{\text{contain}} = \mathbb{E}_{p \sim S_L} [\text{softplus}(d_\theta(p) + m_{in})].$$

A good starting margin is $m_{in} = 0.01L_f$.

Then define pseudo-last support bands:

$$\mathcal{B}_{\text{plantar}} = \{p \in S_L : h(p) \leq \tau_p\},$$

$$\mathcal{B}_{\text{side}} = \{p \in S_L : \tau_p < h(p) \leq \tau_s, |z - c(x)| > 0.7a_t(x)\}.$$

These are the hidden regions you most want to preserve: inner sole, lower sidewall, heel cup.

Now softly anchor the shell near those bands:

$$\mathcal{L}_{\text{band}} = \mathbb{E}_{p \sim \mathcal{B}_{\text{plantar}} \cup \mathcal{B}_{\text{side}}} [|d_\theta(p) + m_b|],$$

where m_b is a small desired shell offset. This prevents the watertight shell from collapsing inward in the unseen sole/interior zone.

1.6.3 Then use the pseudo-last to classify which shell areas should stay active in mSDF

G-Shell’s key idea is to represent an open surface as an island on a watertight shell via an mSDF. So after the shell exists, use the pseudo-last to decide which shell neighborhoods are **topologically valuable** and should be kept active.

Let q_i be shell vertices and $m_\theta(q_i)$ the learned manifold SDF value on the shell. Define a soft keep probability

$$\hat{t}_i = \sigma\left(-\frac{m_\theta(q_i)}{\beta}\right),$$

where lower mSDF means “keep this shell area active.”

Now create pseudo-labels from visible shoe data and pseudo-last support bands:

$$t_i = \mathbf{1}\left[d(q_i, S_{\text{vis}}) < \tau_{\text{vis}} \quad \vee \quad d(q_i, \mathcal{B}_{\text{plantar}} \cup \mathcal{B}_{\text{side}}) < \tau_L\right].$$

Then train with

$$\mathcal{L}_{\text{keep}} = -\frac{1}{N} \sum_i [t_i \log \hat{t}_i + (1 - t_i) \log(1 - \hat{t}_i)].$$

Add a shell-graph smoothness term to avoid fragmented islands:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|E|} \sum_{(i,j) \in E} w_{ij} |m_\theta(q_i) - m_\theta(q_j)|.$$

The complete regularizer is then

$$\mathcal{L}_{\text{pseudo-last}} = \lambda_c \mathcal{L}_{\text{contain}} + \lambda_b \mathcal{L}_{\text{band}} + \lambda_k \mathcal{L}_{\text{keep}} + \lambda_s \mathcal{L}_{\text{smooth}}.$$

In practice, I would use a curriculum:

- **early:** larger λ_c, λ_b so the shell forms around the last and visible outside;
- **mid:** increase λ_k so the mSDF preserves support-related islands;
- **late:** decay all pseudo-last weights and let image evidence dominate where it exists.

This is directly analogous in spirit to clothed-human reconstruction pipelines that use a parametric body model to regularize unseen regions without forcing the final surface to equal the template. PaMIR conditions an implicit field on a parametric body model; ICON uses an estimated SMPL body to guide occluded normal prediction and implicit reconstruction; SeSDF explicitly deforms an SMPL-X-derived SDF to recover the clothed surface.

1.7 Failure cases and validation

The most common failure mode will be **bad foot-in-shoe alignment**. If the aligned SUPR foot is too far forward, your toe allowance becomes meaningless; if it is too low or too wide, the pseudo-last will overconstrain the shell and hallucinate impossible interior volume. This should be checked before any lofting or SDF conversion.

The next failure mode is **using support width as if it were actual inner width**. Many shoe meshes expose outsole/support shape more reliably than the interior cavity, so directly using the raw support width can overestimate the true inner width. That is why I recommend the η_s clamp and why the pseudo-last should be a **soft prior**, not a hard cavity template.

There are also failure modes from over-sculpting:

- too much toe merge makes the forefoot clownish;
- too much arch preservation makes flat sneakers unrealistically hollow;
- too much heel shrink creates self-intersections or an implausibly narrow backpart;
- too much top inflation makes the shell over-thick and weakens visible-view supervision;
- insufficient profile smoothing causes slice-to-slice ripples and unstable SDF signs.

Population variation is a quieter but real failure source. Large scan studies and methodological reviews show foot-shape variation across sex and regions, so one single fixed setting will not behave equally well across all shoes and target feet. That is another reason to keep all allowances normalized and learnable/tunable.

For validation, I would use a small but strict battery.

Geometry diagnostics

- **Plantar RMSE to footbed**

$$\text{RMSE}_{\text{plantar}} = \sqrt{\frac{1}{|S_b|} \sum_{(x,z) \in S_b} (y_b(x,z) - B(x,z))^2}.$$

- **Arch lift profile** along the centerline:

$$a_{\text{arch}}(x) = B(x, c(x)) - y_b(x, c(x)).$$

- **Support conformity ratio**

$$r_w(x) = \frac{2a_t(x)}{w(x)}.$$

This should stay below your chosen η_s cap.

- **Profile roughness**

$$R_a = \sum_i (\Delta^2 a_i)^2, \quad R_H = \sum_i (\Delta^2 H_i)^2.$$

Sudden spikes indicate loft instability.

Topology diagnostics

- the mesh must be watertight,
- no self-intersections,
- exactly one connected toe-box component in slices after s_{box} ,
- stable slice ordering and no contour flips.

Containment diagnostics

- fraction of pseudo-last surface points lying inside the coarse outer shoe envelope,
- histogram of shell-to-last distances in the plantar and heel bands,
- color map of clearance over the pseudo-last surface.

Visualization diagnostics

The single most useful visualization is a stack of sectional overlays every 5% of foot length. Plot, in each slice:

- raw SUPR section,
- toe-merged/smoothed section,
- pseudo-last section,
- support interval $[z_L, z_R]$,
- local footbed B .

If those overlays look right, the 3D mesh almost always looks right. If they do not, no amount of mesh smoothing will save you.

A broader caveat from the footwear-fit literature is that there is still no universally accepted single metric of shoe fit, so section diagnostics plus explicit dimension/error plots are more trustworthy than a single scalar “fit score.”

1.8 Prior art and reading list

The most relevant **shoe-last and foot-to-last** papers for your project are *A Deformation Method for Shoe Last Customization* by Leng and Du, which uses aligned foot/last distance maps; *A CAD Approach for Designing Customized Shoe Last* by the same authors, which explicitly adjusts foot data to toe spring and heel height before last formation; *Shoe-last Design Innovation for Better Shoe Fitting* by Luximon and Luximon; *3D Foot Scan to Custom Shoe Last* by Rout and colleagues, which sections the foot at equal percentage intervals; *Computer Aided Design and Development of Customized Shoe Last* by Sambhav and colleagues; *Customization of shoe last based on 3D design process with adjustable 3D ease allowance* by Mishra and colleagues; and *A Feature-based Shoe-last Design and its Automatic Model Construction* by Li and colleagues. Those papers are the clearest direct precedents for a section-based, parameterized pseudo-last generator.

For **anatomical foot priors and foot-shape variability**, the most relevant sources are *SUPR: A Sparse Unified Part-Based Human Representation*; *Dynamic foot morphology explained through 4D scanning and shape modeling*; *Analysis of 1.2 million foot scans from North America, Europe and Asia*; and the 2024 review *Foot shape assessment techniques for orthotic and footwear applications*. Those provide the strongest justification for using SUPR as an anatomical base while still smoothing it into a last-like envelope.

For **template/SDF priors in reconstruction**, the core references are *DeepSDF*, *Occupancy Networks*, *NeuS*, and *Ghost on the Shell*. Together they give you the right conceptual ladder: continuous signed fields, continuous occupancy, multiview neural surface reconstruction, and finally watertight-shell-plus-mSDF topology control for open/non-watertight geometry.

For the **SMPL/clothed-human analogy** that is closest to your proposed use of a pseudo-last, I would read *PIFu*, *PaMIR*, *ICON*, and *SeSDF*. The common theme is highly relevant to your shoe problem: use an anatomical/template prior to stabilize geometry in hidden or weakly observed regions, but learn a flexible implicit field that is free to deviate where the observed surface demands it.

The practical bottom line is that your pseudo-last should be implemented as a **section-lofted, footbed-aware, toe-merged, heel-controlled, watertight support prior**. If you do that, it becomes a very natural supervisory object for both shell SDF containment and G-Shell mSDF keep/cut decisions in the exact regions where turntable images give you the least information.

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